

Practice Quiz — Market Sensing

Part A: Multiple Choice

1. In Active Inference market research, the primary goal is to:

A) Collect as much data as possible about the market B) Confirm that customers want the product you have designed C) Generate observations that test predictions from your generative model, focusing on prediction errors D) Identify the largest possible total addressable market

1. Why does asking about past behavior produce higher-precision evidence than asking about future intentions?

A) People have better memories of the past than predictions of the future B) Past behavior reflects actual free energy minimization decisions, while stated intentions reflect self-models that may be inaccurate C) Regulatory requirements mandate the use of behavioral data D) Past behavior data is cheaper to collect than intention data

1. An MVP (Minimum Viable Product) is best understood in Active Inference terms as:

A) The cheapest possible version of your product B) A perception instrument designed to generate the highest-precision evidence about specific market hypotheses C) A product with the minimum number of features needed for sale D) A prototype that investors can evaluate

1. When analyzing competitor patent filings as intelligence, you are primarily:

A) Checking whether your invention infringes on their intellectual property B) Inferring their generative model of future technology and market demand from their active states C) Estimating their R&D budget D) Identifying potential licensing opportunities

1. The failure of traditional market research to predict demand for genuinely novel inventions (like the iPhone) is best explained by:

A) Market research firms are incompetent B) Customers cannot provide evidence about needs they have not yet experienced, meaning existing models lack the structure to predict novel

categories C) Novel inventions are always too expensive for market research participants D) Focus groups are unreliable for all product categories

1. Which of the following signals would provide the earliest warning of an impending regulatory change?

A) Publication of the final regulation in the Federal Register B) Academic publications analyzing the technology governance implications of your invention's domain C) News articles about the regulation being enforced D) Your competitor's press release about their compliance strategy

1. In trend detection, signal-vs-noise discrimination is achieved by:

A) Ignoring all sources except major industry analysts B) Testing whether multiple independent signals converge on the same predictive model C) Waiting until the trend is obvious before responding D) Relying on your industry experience and intuition

Part B: Short Answer and Analysis

1. You are developing a new food technology product (e.g., lab-grown meat, novel protein source). Design a sensing strategy that addresses three distinct perception challenges: (a) customer acceptance, (b) regulatory approval trajectory, and (c) competitive landscape. For each, specify the sensing method, the precision level, and the key prediction being tested.

2. Netflix's transition from DVD to streaming involved sensing a signal (growing streaming demand) within a noisy environment (fluctuating DVD revenue). Explain how the Active Inference framework helps distinguish between signal and noise in this case. What specific evidence would have increased confidence that streaming was a genuine trend rather than a temporary phenomenon?

3. You have conducted 15 customer discovery interviews and found that 12 customers confirmed your value proposition while 3 expressed significant concerns about a specific aspect of your invention. Using Active Inference precision weighting, explain how you should interpret these results. Should you proceed with your current model, or

does the minority dissent warrant model revision? What additional sensing would help resolve the ambiguity?